

Warm-up 1

Give an example of a 5×5 nilpotent matrix satisfying the polynomial equation $x^5 = 0$. Can you give an example that has rank 3? What about one that has rank 2?

During class 1

Is this funny? If so, why?

$$\begin{pmatrix} \text{Jordan} & 1 & 0 & 0 \\ 0 & \text{Jordan} & 1 & 0 \\ 0 & 0 & \text{Jordan} & 1 \\ 0 & 0 & 0 & \text{Jordan} \end{pmatrix}$$